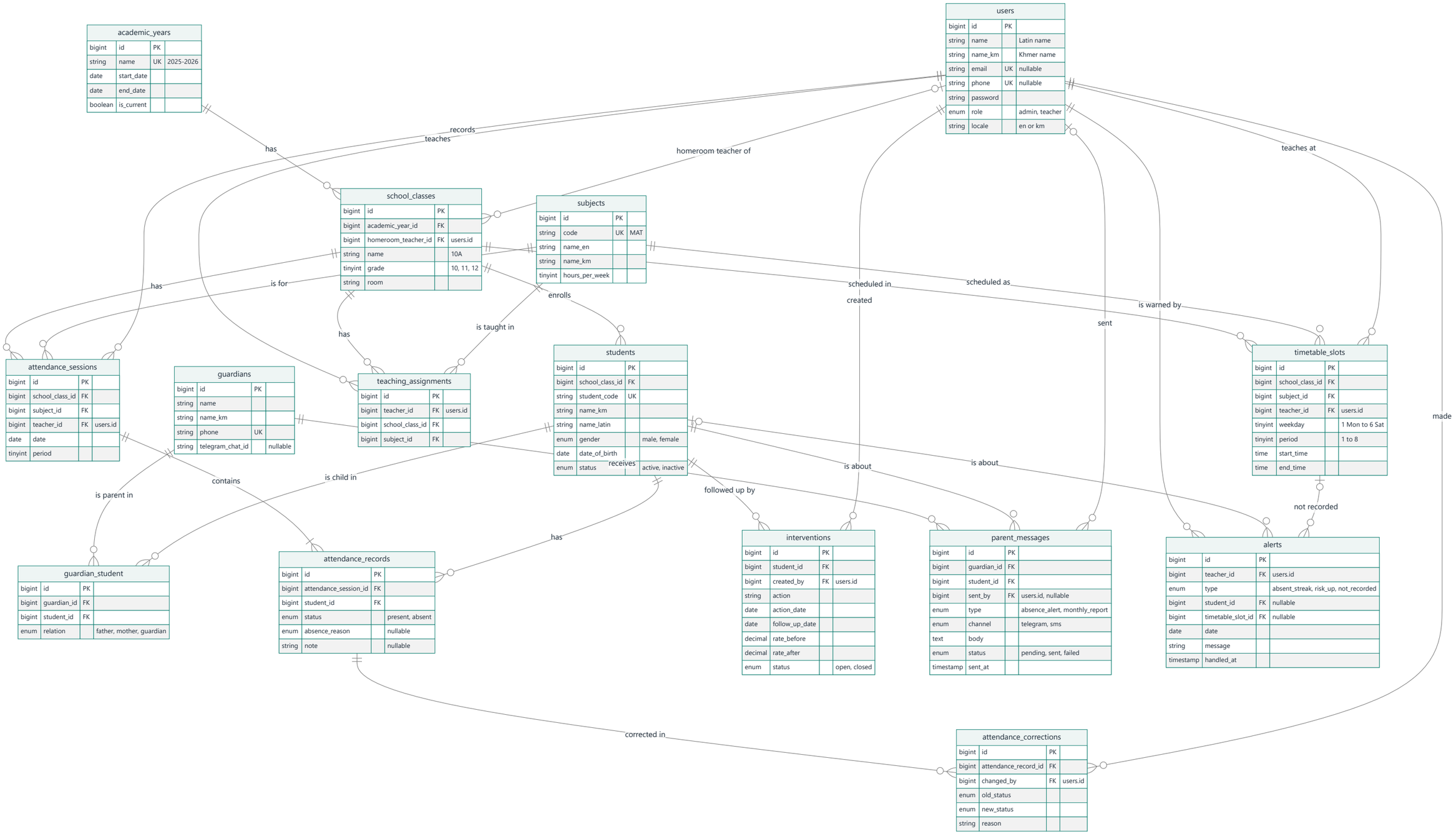
1. ERD: the project tables

Student Attendance Analytics System · Plan v2.0 · Laravel + MySQL · database **attendance_db**



PK primary key **FK** foreign key **UK** unique **||** exactly one **|o** zero or one **o{** zero or many **|{** one or many Every table also has **created_at** and **updated_at**.

2. Data dictionary: school data

Users, the school structure, students and their parents

Plan v2.0 rules built into the database: only Admin and Teacher log in · parents are contacts that receive Telegram or SMS messages · attendance is only Present or Absent, with an optional absence reason.

users

Column	Type	Description
id	bigint	Primary key
name	varchar	Name in Latin letters
name_km	varchar	Name in Khmer
email	varchar	Unique, optional
phone	varchar	Unique, optional; can be used to log in
password	varchar	Hashed (bcrypt), never plain text
role	enum	admin or teacher
locale	varchar	Preferred language: en or km
email_verified_at	timestamp	Laravel default, optional
remember_token	varchar	For "Remember me"

Only school staff have accounts. The role decides which screens a user can open.

academic_years

Column	Type	Description
id	bigint	Primary key
name	varchar	Unique, e.g. 2025-2026
start_date	date	First school day
end_date	date	Last school day
is_current	boolean	Only one year is current

school_classes

Column	Type	Description
id	bigint	Primary key
academic_year_id	bigint	FK → academic_years.id
homeroom_teacher_id	bigint	FK → users.id, optional
name	varchar	e.g. 10A
grade	tinyint	10, 11 or 12
room	varchar	Room number

Unique: (academic_year_id, name). Named school_classes because "class" is a reserved word in PHP.

subjects

Column	Type	Description
id	bigint	Primary key
code	varchar	Unique short code, e.g. MAT
name_en	varchar	English name
name_km	varchar	Khmer name
hours_per_week	tinyint	Lessons per week

students

Column	Type	Description
id	bigint	Primary key
school_class_id	bigint	FK → school_classes.id
student_code	varchar	Unique, e.g. S-10A-001
name_km	varchar	Khmer name
name_latin	varchar	Latin name
gender	enum	male, female
date_of_birth	date	Optional
status	enum	active, inactive (left the school)

Students do not log in. They are records managed by the admin.

guardians

Column	Type	Description
id	bigint	Primary key
name	varchar	Latin name
name_km	varchar	Khmer name
phone	varchar	Unique; used for SMS
telegram_chat_id	varchar	Set when the parent joins the school's Telegram bot

Parents are contacts only. Telegram if they joined the bot, otherwise SMS.

guardian_student

Column	Type	Description
id	bigint	Primary key
guardian_id	bigint	FK → guardians.id
student_id	bigint	FK → students.id
relation	enum	father, mother, guardian

Pivot table: one parent can have many children, one child can have many parents. Unique: (guardian_id, student_id).

teaching_assignments

Column	Type	Description
id	bigint	Primary key
teacher_id	bigint	FK → users.id (role teacher)
school_class_id	bigint	FK → school_classes.id
subject_id	bigint	FK → subjects.id

Who teaches which subject in which class. A teacher sees only these classes. Unique: (school_class_id, subject_id).

timetable_slots

Column	Type	Description
id	bigint	Primary key
school_class_id	bigint	FK → school_classes.id
subject_id	bigint	FK → subjects.id
teacher_id	bigint	FK → users.id
weekday	tinyint	1 Monday … 6 Saturday
period	tinyint	Lesson number, 1 to 8
start_time / end_time	time	e.g. 07:00 to 08:00

The weekly timetable. Gives the teacher "today's schedule". Unique: (class, weekday, period) and (teacher, weekday, period), so nobody is in two places at once.

3. Data dictionary: attendance, analytics and parent contact

Lessons, records, the audit log, follow-ups, parent messages and early warnings

attendance_sessions

Column	Type	Description
id	bigint	Primary key
school_class_id	bigint	FK → school_classes.id
subject_id	bigint	FK → subjects.id
teacher_id	bigint	FK → users.id (who recorded it)
date	date	Day of the lesson
period	tinyint	Lesson period, 1 to 8

One lesson where attendance was taken. Unique: (school_class_id, date, period), so the same lesson cannot be recorded twice.

attendance_records

Column	Type	Description
id	bigint	Primary key
attendance_session_id	bigint	FK → attendance_sessions.id
student_id	bigint	FK → students.id
status	enum	present or absent, nothing else
absence_reason	enum	Optional: not_given, sick, family_matter, school_activity, other
note	varchar	Optional

One row per student per lesson. Unique: (attendance_session_id, student_id). All analytics come from this table. Rate = present ÷ all records.

attendance_corrections

Column	Type	Description
id	bigint	Primary key
attendance_record_id	bigint	FK → attendance_records.id
changed_by	bigint	FK → users.id (who made the change)
old_status	enum	present or absent, before
new_status	enum	present or absent, after
reason	varchar	Why it was changed
created_at	timestamp	When the change was made

Audit log. Teachers correct on the same day; the admin corrects past days. Records are never changed silently.

interventions

Column	Type	Description
id	bigint	Primary key
student_id	bigint	FK → students.id
created_by	bigint	FK → users.id
action	varchar	e.g. Parent message (Telegram), meeting with parent
notes	text	Optional
action_date	date	When the action was taken
follow_up_date	date	When to check again
rate_before / rate_after	decimal	Attendance rate before, and when closed
status	enum	open or closed
closed_at	timestamp	When it was closed

Follow-up actions for at-risk students. rate_before and rate_after show whether the action helped.

parent_messages

Column	Type	Description
id	bigint	Primary key
guardian_id	bigint	FK → guardians.id
student_id	bigint	FK → students.id
sent_by	bigint	FK → users.id; empty = sent by the system
type	enum	absence_alert or monthly_report
channel	enum	telegram or sms
body	text	Message text, Khmer then English
status	enum	pending, sent, failed
error	varchar	Why sending failed
sent_at	timestamp	When it was delivered

Every message to a parent, so the school can prove what was sent and when.

alerts

Column	Type	Description
id	bigint	Primary key
teacher_id	bigint	FK → users.id (who should see it)
type	enum	absent_streak, risk_up, not_recorded
student_id	bigint	FK → students.id, for student alerts
timetable_slot_id	bigint	FK → timetable_slots.id, for "not recorded"
date	date	Day of the alert
message	varchar	e.g. Absent 3 school days in a row
handled_at	timestamp	When the teacher handled it
handled_by	bigint	FK → users.id

Early warnings. Risk levels themselves are calculated, not stored: Critical (<70% or 5+ days in a row), High (70–79%, 3–4 days or down 10+ points), Medium (80–89% or down 5–9), Low.